Planet-Scale Energy Yield Potential of Next-Generation Bifacial, Multiterminal, Perovskite-Silicon Tandem Solar Farms

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***Abstract*—To continue reducing the levelized cost of solar energy, the photovoltaics (PV) industry is developing higher efficiency perovskite-based tandem solar cells. Among the various options, the two-terminal (2T) tandem has traditionally garnered the most interest and is expected to enter the market soon. However, the bifacial 2T perovskite-silicon (PVK-Si) tandem cell, constrained by current-matching requirements, would offer diminished energy gains in large-scale solar farms, especially when subjected to suboptimal albedo conditions. The 3/4T tandems obviate current matching and are expected to outperform 2T-tandem cells. However, the actual location-specific yield potential and relative gain of bifacial 3/4T tandems has not been reported in the literature. In this work, we use a novel end-to-end, multiscale simulation framework to carry out the first planet-scale simulation of single-axis-tracking solar farms employing *bifacial* PVK-Si 3/4T tandem in various ground albedo conditions. The analysis shows that the 3/4T cells offer up to 5% and 23% mean increase in annual energy yield compared with 2T-tandem and single-junction heterojunction solar cells in Earth's average albedo ($R_A = 30\%$). Importantly, unlike the 2T tandem, the 3/4T tandem maintains its performance advantage across a wide range of albedo conditions, enabling flexible subcell design. The findings should encourage further research efforts aimed at tackling the recognized challenges associated with 3/4T technologies, such as minimizing optical losses and scaling up cell-to-module processes, to fully realize the potential of PVK-Si tandem technology.**

***Index Terms*—Modeling, multiterminal, perovskite-silicon (PVK-Si) solar cells, simulation, solar farms, tandem solar cells.**

I. Introduction

TO FURTHER reduce the levelized cost of solar energy (LCOE), the photovoltaics (PV) industry is developing next-generation high-efficiency module technologies, such as monofacial and bifacial perovskite-silicon (PVK-Si), perovskite–perovskite, and perovskite-organic (PVK-BHJ) tandem solar cells. Among these options, PVK-Si tandems are particularly attractive due to the potentially cheap integration of vapor- or solution-based PVK with commercially mature monofacial and bifacial Si solar cell technologies. The efforts have been driven by two key observations: the thermodynamic promise of remarkable efficiency gain for tandem solar cells [1], [2] and the fortuitous near-perfect (i.e. thermodynamically optimum) bandgap matching of the three-dimensional (3-D) and 2-D/3-D heterostructure perovskite [3] and silicon-heterojunction (Si-HIT) solar cells [4] with a typical average ground surface albedo (R_A) of 30% across the world. With durable, low-cost solution-processed 2-D/3-D perovskite fabrication methods on the horizon [3], the PVK/Si tandem solar cells appear to be the likely candidate technology for next-generation utility-scale solar farms. However, the LCOE is location-specific: the thermodynamic promise of high-efficiency PVK-Si cells must be tested under realistic operating conditions across the world.

In the thermodynamic limit with perfectly matched subcell currents, two-terminal (2T) and four-terminal (4T) tandem stacks must necessarily have comparable efficiencies (~45.5% under terrestrial AM 1.5 global radiation) [5], [6]. Even with realistic optical and electrical losses, the bifacial double-junction PVK-HIT tandem is expected to have ~33% normalized efficiency under the AM 1.5G spectrum [4]. The conditions in an outdoor solar farm, however, differ considerably compared with the thermodynamic or laboratory setup. In practice, solar modules are arranged in closely packed solar arrays experiencing location and time-dependent variations in irradiance and temperature. *Under such conditions, the conclusions derived from the thermodynamic analysis are no longer valid.* For example, the analysis by Patel et al. [7] showed that the energy gain of bifacial 2T PVK-HIT [see Fig. 1(a)] in a solar farm would be constrained by the current-matching requirement in its subcells. This is because, for a bifacial 2T-tandem cell deployed in a solar farm, the **effective** albedo (R_A^*), defined as the ratio of irradiances on the rear and front planes of the array, i.e.,

$$R_A^* = \frac{I_{\mathrm{rear}}}{I_{\mathrm{front}}}$$

determines the current-matching condition and yields performance benefits. The gain of the bifacial 2T tandem (26%

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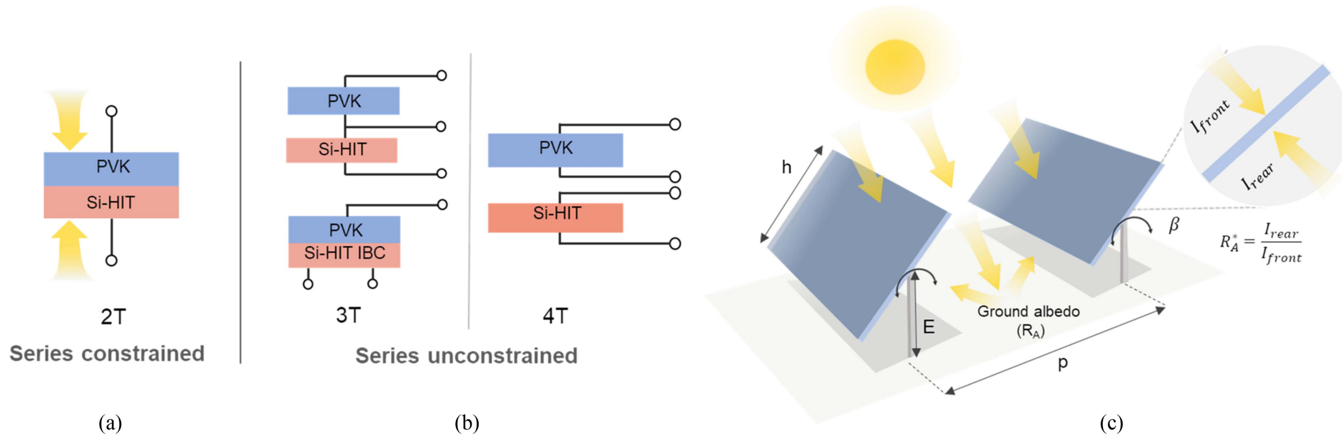


Fig. 1. (a–b) Three possible bifacial PVK-Si bifacial tandem solar cell configurations. For the 2T configuration, the subcells connected in series and, thus, currents must match. (b) This constraint is obviated in 3T and 4T configurations by introducing additional terminals. (c) Schematic diagram of a bifacial solar farm. Bifacial modules collect additional light reflected from the ground surface. Various important system design parameters, such as elevation (E), ground surface albedo (R_A), effective albedo (R^*_A), and row tilt angle (β) are indicated on the diagram.

efficiency cell) compared with single-junction bifacial HIT (25% efficiency) was found to be only 4%–5% globally only when the in-array R^*_A was close to its design R^*_A required for current matching. It was also shown that under nonoptimal albedo conditions (e.g., due to seasonal variation), a 2T-tandem cell may fail to outperform a high-efficiency HIT cell. Since the in-array R^*_A will always deviate from a cell's design R^*_A, such sensitivity to albedo conditions calls into question the viability of bifacial 2T tandem as the optimal candidate technology for the next generation of bifacial PV systems across the world. Fortunately, the current-matching requirement is obviated by electrically decoupling the subcells in a three or four [3/4T, see Fig. 1(b)] tandem configuration [8]. The global analysis presented in this article shows, *for the first time,* that 3/4T configurations promise to unleash the potential of PVK-Si tandem technologies in the context of utility-scale solar farms, provided that the industry can address key challenges associated with cell and module technologies.

Both traditional 2T and multiterminal (3/4T) architectures have been extensively discussed in the literature. While historically multiterminal and multijunction cells have been explored for space applications with well-documented performance gain [9], for terrestrial applications, the academic and industrial research has focused on 2T tandem, owing to its monolithic fabrication and easier system integrability. This interest has culminated in the notable commercialization efforts of PVK-based solar cells by several companies across the world, e.g., Oxford PV (PVK-Si-HIT tandem), Microquanta (all PVK), etc. Recently reported monofacial 2T-tandem cell with 31.25% (1 cm^2 cell area) efficiency has further increased the interest in this technology [10]. At the same time, monofacial 4T-tandem cells have been demonstrated to have efficiencies reaching 27.7% (1 cm^2 cell area) [11]. Anticipating the commercialization of 2T tandems, the interest has also yielded several theoretical [12], [13], [14], [15], [16] and experimental studies [17], [18], [19] investigating the outdoor performance of PV systems employing these cells.

Utilizing the ground-reflected light, a single-junction, high-efficiency, bifacial cell is likely to maintain its advantage over the monofacial tandem cells in terms of energy yield [4]. Thus, to compete with a rapidly growing bifacial market and reduced cost/unit, future tandem would have to be bifacial. In contrast to monofacial cells, bifacial cells can collect irradiance through the rear face, and this results in a higher energy yield per area. Thus far, only a handful of groups have reported *bifacial* 4T PVK-Si tandem cells. Coletti et al. [20] reported the first bifacial 4T device with 30.5% output power density normalized to 1 Sun with 0.2 Suns rear irradiance in the lab. In a small-scale outdoor experiment with the same cell, they recorded a 20% higher yield with bifacial minimodules compared with monofacial ones under ∼20%–40% albedo condition. In another study, Kim et al. [21] reported a 30% efficiency bifacial 4T-tandem cell under 0.5 Sun rear irradiance measured in the laboratory. More recently, Manshanden et al. [22] reported outdoor measurements with 100 cm^2 bifacial 4T minimodules showing that the bifacial gain can be over 26% that of monofacial for 10% albedo. Their study also found that the ratio of irradiances on the front and the rear face is important in characterizing the performance benefits of bifacial tandem cells.

Since the monolithic 4T configuration suffers from optical losses associated with the additional contacts in between the cells, a 3T configuration may be preferred by being monolithically integrable with lower optical losses and not requiring current matching between subcells. In a theoretical study, Santbergen et al. [23] found that 3T-tandem efficiencies of up to 32% can be obtained. Recently, Tockhorn et al. [24] demonstrated the first monofacial 3T PVK-Si tandem cell with interdigitated back-contact (IBC) having 17.1% experimental efficiency. Their simulation demonstrated a potential efficiency of 27%. Regarding improved energy yield potential, a theoretical study by Gota et al. [25] showed that the monofacial 3T architecture can deliver 3%–9% energy yield compared with monofacial 2T in different locations across the US. Their analysis also finds that 4T architecture, despite its relaxed design constraints in terms of PVK bandgap and thickness, fails to demonstrate a yield advantage compared with an optimally designed 2T tandem, owing to parasitic optical losses. Indeed, a recent worldwide technoeconomic analysis also finds 3T and voltage-matched modules to be the most promising tandem configuration [26]. By optimally designing the gaps in the bottom-cell's IBC, a *bifacial* 3T tandem can be realized to further increase the power output [27]. However, 3T tandems still come at a cost of increased interconnection complexity and/or the need to use a relatively more expensive Si IBC bottom cell. In this regard, mechanically stacked 4T PVK-Si tandem modules may prove to be an economically viable alternative. Therefore, regardless of the cell architecture, the additional cost of high-efficiency *bifacial,* 3/4T-tandem cells can be justified only if they significantly outperform their constituent single-junction cells under realistic location-specific conditions.

Most yield performance analyses in the literature have thus far concentrated on **monofacial** 3/4T-tandem cells. As such, the yield potentials of bifacial 3/4T cells remain unknown under realistic farm conditions. While modeling the outdoor energy yield potential of a technology, often the multi-row characteristic of a solar farm is neglected for simplicity. However, when evaluating the yield potential of a bifacial solar farm, one must carefully account for the critical impact of row-to-row shading in determining the irradiance available to the rear surfaces of the modules. The modules near the center of a multirow bifacial array receive lower rear irradiance, owing to the shadow cast by the adjacent modules and rows. Since the reduction in rear irradiance can be up to 15% compared with a standalone single row of modules [28], [29], the performance of a single-row system will overestimate the potential gain. Furthermore, the analyses have often been limited to small geographical regions that are unable to inform global utility-scale performance expectations.

In this work, we address the aforementioned gaps by elucidating the annual energy yield potentials of solar farms across the planet when employing *bifacial, multiterminal* tandem cells under realistic time-varying irradiance and temperature. Toward this goal, we developed a detailed and experimentally validated physical model of the multirow solar farm to carry out global-scale simulations of solar farms employing bifacial 3/4T-tandem solar cells to answer the following questions of broad technological relevance.

1) Given the spatiotemporal variation in irradiance and temperature across the world, what is the worldwide annual energy yield potential of solar farms employing 3/4T tandems?
2) What are the expected relative gains of 3/4T-tandem farms in comparison with 2T tandem and its constituent single-junction Si-HIT cells?

The rest of this article is organized as follows. Section II summarizes our modeling framework. Section III discusses the worldwide energy yield potential and relative gains compared with 2T and single-junction HIT cell. Section IV discusses the performance of a single farm to elucidate the role of effective albedo in determining tandem benefits. Finally, Section V concludes this article.

II. Modeling Framework

We have used PV-MAPS [30], a sophisticated and experimentally validated solar farm model, to carry out global simulations of tandem solar farms. The simulation framework has three components: the irradiance model; the electrical model; and the temperature model. The electrical and temperature models were generalized to include the specialized physics of PVK-HIT tandem cells. The PV-MAPS framework has been described in detail in [7]; here, we summarize its key components.

The irradiance model computes the solar radiation collected by a solar farm on a given day at any location on Earth. First, for any given location, the PVLIB toolbox [31] from Sandia National Laboratory is used to compute the daily sun path and the clear-sky global horizontal irradiance (GHI). Then, the clear-sky GHI is scaled by the observed GHI values from the NASAs climatological normal dataset [32] to obtain the local GHI profiles. The computed local GHI is then decomposed into direct irradiance (DNI) and diffuse irradiance (DHI) components using the Orgill–Hollands–Perez model [33]. These irradiance components constitute the type and measure of solar resources available at a location. Next, the model computes the total irradiance collected by the array whose design is described through a set of parameters. Fig. 1(c) illustrates the relevant solar farm design parameters, such as module height (h), module angle (β), azimuth angle (γ_A) from the north, elevation from the ground (E), pitch (p), and ground surface albedo (R_A). With these parameters defined, the model calculates the spatial distribution of sunlight incident on the module surfaces and the ground. Considering rows of the solar farm to be of infinite lengths, the model calculates the shadows cast by the rows according to the position of the sun. Assuming isotropic diffuse radiation, the view-factor method is employed to calculate the diffuse light reaching the module surfaces from both the sky and the ground below. These calculations are carried out for every 1-min time step. To simulate a farm employing a horizontal single-axis sun tracker, the module tilt angle about the rotation axis is updated at each time step to maximize the power output. In this analysis, we consider tracking systems with a north–south-oriented axis.

Next, to determine the electrical output of the bifacial 3/4T tandem, we first calculated its subcell optical characteristics and efficiency as a function of the front (I_{front}) and rear irradiances (I_{rear}) at a fixed temperature (25 °C). At the cell level, a full-wave optical model implemented by the transfer-matrix approach gives absorption and photocarrier generation profiles within the single-junction or tandem solar cell for a given front and rear irradiance [4], [34]. Fig. 2(a) illustrates an example when $I_{\text{front}} = 1$ Sun and $I_{\text{rear}} = 0.3$ Suns. Here, we assume the global standard (AM 1.5G) spectrum for both I_{front} and I_{rear} with wavelengths ranging between 300 and 1500 nm. Neglecting location-specific spectral irradiance may lead to a biased estimate of yield potential. Ripalda et al. [35], analyzing monofacial, double-junction cells with slightly different bandgaps, estimated the impact on yearly yield potential in continental USA to range from -3.4% to -1.1% for 2T cells and -2.2% to 1.5% for 4T cells. For bifacial cells, the surface-specific spectral albedo would have to be accounted for as well. As such, the yield

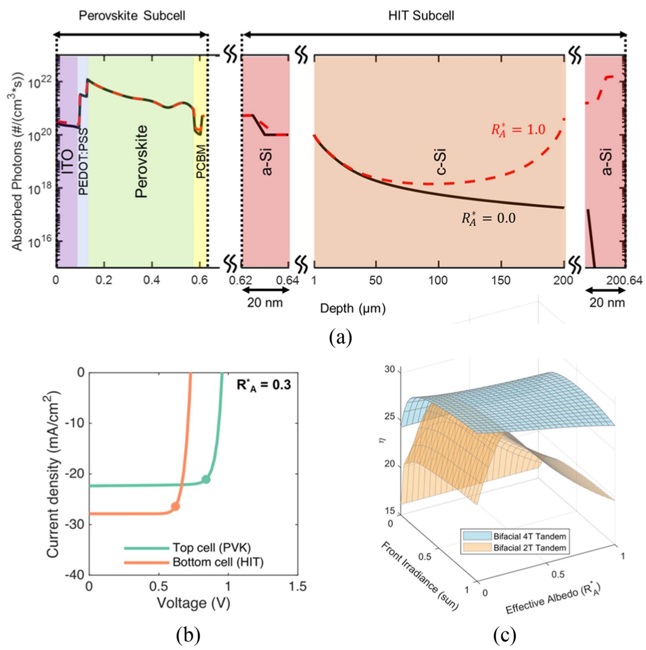


Fig. 2. Simulation of 3/4T-tandem cells. (a) Photogeneration profile of the bifacial cell at 1 Sun (=1000 W·m^{-2}) front irradiance (from left) and 1 Sun (0 Sun) rear irradiance (from right) shown by dashed red (solid black) line. Both front and rear irradiances were assumed to be AM 1.5G in this analysis. (b) *J–V* characteristics of the subcells in a 3/4T-tandem cell under 1 Sun front and 0.3 Suns rear irradiance. The circles indicate the maximum power points for each subcell, highlighting the flexibility of operation in a 3/4T architecture. (c) Efficiency landscapes of 4T and 2T tandems as a function of front irradiance and R_A^*. The efficiency of the bifacial 3/4T tandem remains relatively insensitive to variations in R_A^* and front irradiance, while the efficiency of the 2T tandem is maximized around $R_A^* \approx 0.18$. Deviation from the design albedo during operation leads to current mismatch and rapid reduction in the efficiency of the 2T tandem.

potential estimates presented here for AM 1.5G should be seen as an upper limit estimate, which will serve as a baseline for future refinements and improvements.

Subsequently, the position-resolved transport of photogenerated carrier concentration inside the cell was simulated by a self-consistent solution of Poisson and continuity equations with the device simulator MEDICI [36] to find the corresponding subcell *J–V* characteristics, as shown in Fig. 2(b). The output power of the cell can then be readily calculated: for the 3/4T tandem, the total power output is the sum of the maximum power outputs of each subcell. The optical model neglects the parasitic optical losses inherent in a practical 3/4T device, e.g., owing to the additional intermediate transparent conductive oxide and spacer layer in 4T; or owing to IBC in a bifacial 3T configuration. We assume that these losses will be suitably reduced as the technology matures (or could be effectively included through a derating factor). Additionally, we neglect cell-to-module upscaling losses. By disregarding the optical losses and considering 3T and 4T to be electrically equivalent, we focus on the potential yield benefits of these multiterminal architectures under the best-case scenario. The practical gain would be somewhat lower.

Having computed the subcell *J–V* characteristics and corresponding output power for various combinations of front and rear irradiances, we can construct the efficiency landscape of 3/4T-tandem cells. Fig. 2(c) shows the efficiency landscape of the 3/4T-tandem cell (blue surface) as a function of effective albedo (R_A^*) and front irradiance. As mentioned previously, R_A^* is defined as the ratio of the front and rear irradiances normal to the planes of the cell/array.

In a solar farm, R_A^* depends on the position, time, array geometry, R_A, and surroundings, and directly determines tandem cell efficiency. This is in contrast with R_A, which is considered a time-independent system design parameter representing the reflectance of the ground surface underneath. (The definitions are also listed in Table S1 in the supplementary material.) The efficiency landscape of the 2T tandem is also shown in Fig. 2(c) (brown surface). The two bifacial cell configurations have easily distinguishable efficiency landscapes. The landscape of the 3/4T tandem is relatively insensitive to changes in R_A^*. By contrast, the 2T-tandem's efficiency is maximized only for a particular R_A^* (determined by its subcell's bandgap and thickness) and diminishes rapidly under different circumstances. Only if the current-matching conditions are met, which in this case when $R_A^* \sim 0.18$, the efficiencies of the 3/4T tandem and 2T tandem are equal.

Having determined the efficiencies, we determined the effect of location-specific temperature variations by self-consistently computing the cell temperature (T_{cell}) and the temperature-dependent efficiency $\eta(T_{\text{cell}})$ while evaluating the farm's power output. Since each subcell in a tandem cell has individual temperature coefficients (TC), the 3/4T-tandem's *effective* temperature coefficient (TC*) is given by

$$\text{TC}^* = \frac{I_{\text{mp,PVK}} V_{\text{mp,PVK}}\ \text{TC}_{\text{PVK}} + I_{\text{mp,HIT}}\ V_{\text{mp,HIT}}\ \text{TC}_{\text{HIT}}}{I_{\text{mp,PVK}} V_{\text{mp,PVK}} + I_{\text{mp,HIT}} V_{\text{mp,HIT}}}.$$

Temperature coefficients for PVK (TC_{PVK}) and HIT (TC_{HIT}) subcells were assumed to be $-0.379\%/\text{K}$ and $-0.258\%/\text{K}$, respectively [7]. (A detailed derivation of the temperature model can be found in the supplementary material.) With TC^*, the temperature-corrected efficiency $\eta(T_{\text{cell}})$ of tandem can be computed as a function of ambient temperature (T_{amb}) and total light collected at the plane of the array (P_{POA}). Finally, having applied the temperature-related corrections, the electrical model accounts for the losses due to partial shading and calculates the power output of the solar farm for each time step. The daily yields are computed for a year to give the yearly energy yield (YY) of the solar farm.

III. Global Yield Potential of Bifacial 3/4T-Tandem Solar Farm

Using the above-described modeling framework, we carried out global simulations of bifacial single-axis-tracking solar farms employing PVK-HIT 3/4T-tandem modules. Given the significant influence of R_A on the energy yield potential of bifacial systems, we computed the yield potentials for three albedo coefficients: $R_A = 0.1$, 0.3, and 0.8. Although these values may not be practical for certain locations on Earth, they allow for a comparison of potential technology-related benefits worldwide, regardless of location-specific ground surface

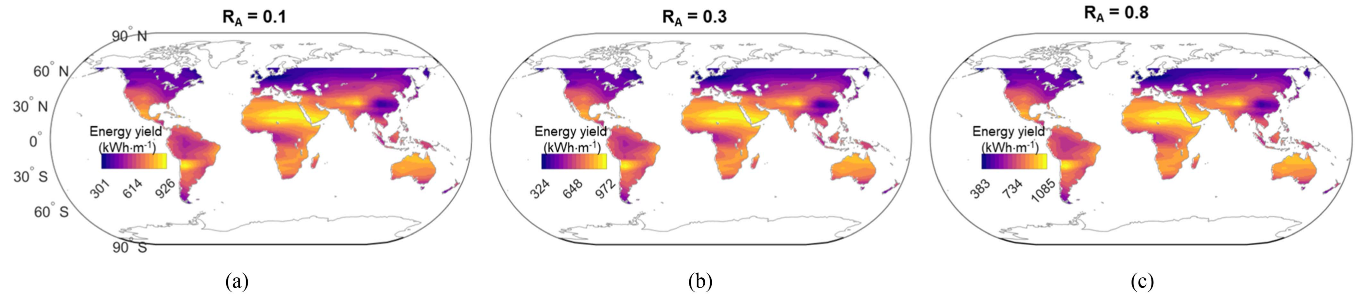


Fig. 3. Global yearly energy yield (YY) potentials of tracking solar farms utilizing 3/4T-tandem modules for different ground albedo conditions. (a) $R_A = 0.1$. (b) $R_A = 0.3$. (c) $R_A = 0.8$. The solar panels are tracked along an east–west-oriented horizontal axis for optimal power output. Higher insolation regions exhibit the maximum yield potentials, aligning with expectations. Also as expected, the energy yield potential of the bifacial farms shows an increase with increasing R_A. The yield potentials were computed assuming standard AM 1.5G spectrum at every location.

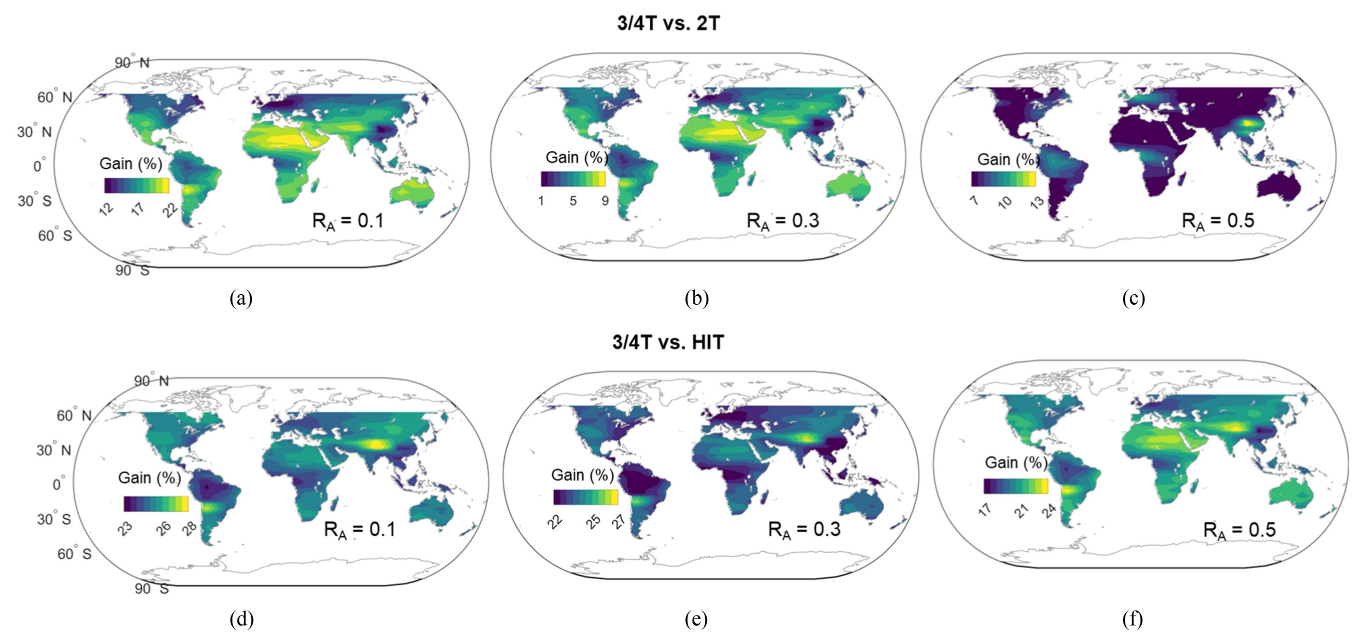


Fig. 4. Global relative gains in YY of 3/4T tracking solar farm compared with 2T (a–c, top-row) and HIT (d–f, bottom-row) tracking solar farms in different albedo conditions. Compared with both 2T and HIT, significant gains are observed across the world for systems employing 3/4T tandem. The gains are especially notable at low albedo condition ($R_A = 0.1$) shown in (a) and (d). The yield potentials were computed assuming standard AM 1.5G spectrum at every location.

properties. Throughout these simulations, we assumed $p/h = 3$ as the optimal system design parameter worldwide [37], which is equivalent to a 33.3% ground coverage ratio. The array's elevation (E) was set to 1 m. The module height (h) was chosen to be 1 m as well [37]. The yearly energy yields (YY) are reported in energy yield in kilowatt hour (kWh) per row length of the farm ($\text{kWh}\cdot\text{m}^{-1}$). In this section, we will illustrate the global trends in the performance benefits of 3/4T-tandem cells based on the results obtained from the global-scale simulations. To explain the observed global trends, Section IV will delve into the daily performance of a solar farm under different R_A.

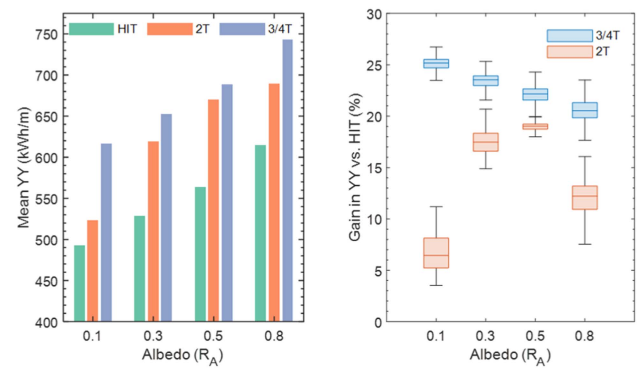
Fig. 3 illustrates the estimated YY of 3/4T solar farms lying within $\pm 60^\circ$ latitudes for the three R_A. In all three cases, regions with higher insolation, such as the Middle East, northern Africa, and the southern United States, exhibit the highest yields, as expected. As one moves away from the equator toward the poles, we observe a nearly threefold reduction in potential yield. When $R_A = 0.1$ [see Fig. 3(a)], the energy yield potential ranges between 301 $\text{kWh}\cdot\text{m}^{-1}$ at high latitudes to 926 $\text{kWh}\cdot\text{m}^{-1}$ near the equator. When $R_A = 0.3$ [see Fig. 3(b)], the yield potential increases everywhere and ranges between 324 and 972 $\text{kWh}\cdot\text{m}^{-1}$. An additional increase in albedo ($R_A = 0.8$) leads to further improved yield potentials with the expected yields ranging between 383 and 1085 $\text{kWh}\cdot\text{m}^{-1}$. The predicted energy yields should be viewed as the upper limits of the 3/4T solar farm performance because the optical loss related to inter-subcell contact or cell-to-module upscaling losses will additionally degrade the performance.

Next, we calculate the relative benefits of 3/4T-tandem cells compared with 2T tandem and HIT in terms of energy yield potential. First, let us compare the yield gains relative to 2T for the three different albedo conditions [see Fig. 4(a)–(c)]. Reviewing at the top row of figures, the advantage of 4T-tandem cells is evident and significant. The maximum relative gain is obtained when $R_A = 0.1$ ranging between 12% and 22%. Higher relative gains are expected in regions near the tropics, such as Saudi Arabia and northern Mexico, where direct insolation is predominant, and the diffuse fraction is lower. These substantial gains occur because, unlike bifacial 2T-tandem cells, the subcells in a 4T tandem are electrically decoupled, and their performance is not limited by the bottom-cell current under low albedo conditions. Being series unconstrained, a 3/4T-tandem cell consistently outperforms a 2T-tandem cell under similar operating conditions by maximally utilizing the incident light. When $R_A = 0.3$, as shown in Fig. 4(b), the relative gain of the 3/4T tandem is reduced to a range between 1% and 9% as 2Ts output improves with increasing R_A. Relatively lower gains ($\sim$1%) are predicted at locations with higher diffuse fractions, such as eastern China and Amazonia. This results from enhanced rear irradiance collection, leading to improved bottom-cell current in these areas due to the prevalence of diffuse light. However, when the albedo is high (R_A= 0.8), the relative gain of 4T-tandem cells increases once again, ranging between 7% and 13%. Additionally, for this R_A, we observe that locations with higher diffuse fractions exhibit *greater* 3/4T gains, opposite of what was observed in the previous two cases. This is because, under such high-albedo conditions, the current in the 2T tandem is primarily limited by the perovskite top cell, and an enhancement in bottom-cell current at these locations leads to a larger current mismatch. With the deterioration in 2T-tandem performance under such high albedo, 3/4T gains increase.

The electrically decoupled subcells of the 3/4T tandem also outperform its constituent HIT cell regardless of albedo condition. Fig. 4(d)–(f) shows the worldwide relative gains of 4T-tandem farms in comparison with the HIT. Again, we observe the 3/4T-tandem farms demonstrating a higher yield potential everywhere: the relative gains span between 23% and 28% when $R_A = 0.1$, 22% and 27% when $R_A = 0.3$, and 17% and 24% when $R_A = 0.8$. For a given albedo, slightly higher relative gains in annual yield are expected for 4T tandem in regions experiencing lower annual average temperatures (e.g., Alaska and northern Canada) due to the improved temperature coefficients of tandems.

The results of the global simulations are summarized in Fig. 5 for various albedo conditions. In Fig. 5(a), the mean yearly yields of solar farms using 2T, 3/4T, and HIT cells are compared for different albedo values ($R_A = 0.1$, 0.3, 0.5, and 0.8). We find that 3/4T tandem promises the highest mean yield across the world followed by 2T and HIT for any albedo condition. The boxplots in Fig. 5(b) show the distributions of 2T and 3/4T-tandems' relative gains compared with Si-HIT in the different albedo conditions. These results demonstrate that the gain of the 2T tandem is highly sensitive to changes in albedo, whereas the 3/4T tandem exhibits significant and relatively stable gains across different albedo conditions.

Due to current-matching requirements, the benefits of the bifacial 2T tandem can only be maximized within a certain albedo range. If the albedo condition is too low (e.g., $R_A = 0.1$) or too high (e.g., $R_A = 0.8$), the performance benefits of the bifacial 2T tandem diminish. On the other hand, the 4T tandem maintains higher gains even with varying albedo conditions, with the gains compared with the 2T tandem particularly increasing at low- and high-albedo conditions. This is a crucial practical consideration since albedo conditions can naturally change throughout the year in certain regions due to seasonal changes. Overall, the relative gains decrease as the albedo increases, as the efficiency of bifacial cells also decreases with higher albedo due to an increase in operating temperature.



(a) (b)

Fig. 5. Summary statistics of the global simulation results for tracking bifacial solar farms at various albedo conditions. (a) Global mean yearly energy yields (YY) of bifacial 3/4T-tandem cells in comparison with 2T and HIT cells. (b) Relative gain in energy yield for 3/4T and 2T tracking farms compared with single-junction HIT cells. The series-constrained 2T tandem demonstrates significant gains within a narrow range of albedo conditions, while the series-unconstrained 3/4T tandem outperforms both 2T and HIT cells by a substantial margin at every albedo condition.

Notably Fig. 5(b) shows that most benefits with the 2T-tandem cell can be obtained across the world when $R_A = 0.5$ even though the highest efficiency is obtained when $R_A^* \approx 0.18$ [see Fig. 2(c)]. This disparity reveals a challenge for bifacial 2T-tandem solar farms: despite designing a cell for a specific R_A^*, the R_A of a farm will have to be higher than its design R_A^* to maximize the benefits. If a cell is tailored for a different R_A^* value, the R_A that produces the maximum output will change, but the general trend, as depicted in Fig. 5(b), will remain consistent. Adding to this challenge, R_A^* varies throughout the day within a farm, depending on its geometry, resulting in varying relative benefits for bifacial tandem cells.

So far, we have discussed the yield performance and gains of the bifacial tandem cells as a function of R_A. In Section IV, we will elucidate the critical role played by R_A^* in determining the performance benefits of bifacial tandem cells.

IV. Effective Albedo in Solar Farms Determines the Bifacial Tandem's Benefits

To elucidate the impact of R_A^* on the benefits of 3/4T tandems, we analyze the daily relative efficiencies of a hypothetical solar farm located in Yuma, Arizona, USA, considering three different ground albedo conditions ($R_A = 0.1$, 0.3, and 0.8). At this location, the YY potential of a 3/4T farm with $R_A = 0.3$ is determined to be 764 kWh$\cdot$m^{-1}, resulting in a 7% gain compared with 2T tandems and a 24% gain compared with HIT cells. For $R_A = 0.8$, the same 7% gain is expected relative to 2T. However, the potential gain compared with 2T rises to 20% under low albedo conditions ($R_A = 0.1$).

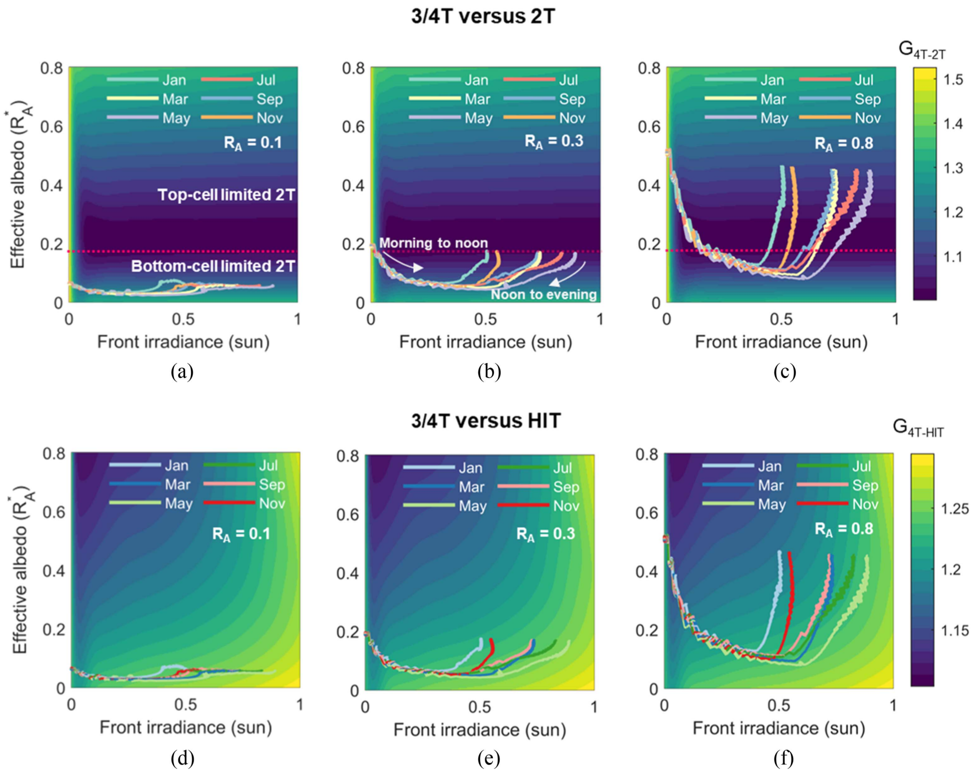


Fig. 6. Daily evolution of 3/4T-tandem's relative gains versus effective albedo and front irradiance compared with (a)–(c) 2T tandem and (d)–(f) Si-HIT for a bifacial tracking solar farm in Yuma, Arizona, US. The tracker's axis is oriented in the north–south direction. The dashed line indicates the R_A^* value (0.18) for which subcell currents match. On a given day of the year, the combination of effective albedo (determined by the ground albedo and the array geometry) and front irradiance determines the gain of 3/4T tandem compared with 2T and HIT. At this location, 3/4T tandem consistently outperforms both 2T and HIT throughout the year. Both front and rear irradiances were assumed to be AM 1.5G.

The figures in Fig. 6 show the relative increase in 3/4T cell efficiency (at 25 °C) as a function of front irradiance and R_A^*. It is worth reminding that, in a solar farm, the R_A^* is defined as the ratio of front and rear plane-of-array irradiances taking into consideration the time-varying array geometry and the sun's position at the location. In Fig. 6, each figure's background shows the 3/4T-tandem's relative efficiencies (G) (e.g., $G_{4T-2T} = \eta_{4T}/\eta_{2T}$, $G_{4T-HIT} = \eta_{4T}/\eta_{HIT}$) as a function of R_A^* and I_{front}. At a particular time, the operating G of 3/4T, a farm will be determined by the R_A^* and I_{front} collected by the array at that time. We observe from the color bar that G is always greater than 1 compared with 2T or HIT for any combination of R_A^* and I_{front}. This is expected from the positive gains observed in global maps, as shown in Section III. The dashed line on the figures in the top row indicates $R_A^* = 0.18$ when the subcell currents match.

Now, let us consider Fig. 6(b), which demonstrates the relative farm performance when $R_A = 0.3$. On a given day, both I_{front} and R_A^* vary throughout the day; each overlaid trace shows the 3/4T relative efficiencies during a representative day from different months. (See Fig. S1 in supplementary materials for an example of R_A^* versus time.) On any day, the I_{front} collected by solar farms is minimum in the mornings and the afternoons but maximized near midday when the sun reaches its highest zenith. Since the R_A^* is defined as I_{rear}/I_{front}, it is maximized when I_{front} is minimized. Therefore, higher R_A^* can be observed when I_{front} is lower. This occurs during early mornings and evenings. As the day progresses, I_{front} increases and the tilt angle of the array is adjusted to maximize the power output. We notice R_A^* increasing as I_{front} increases. For an east–west tracking system, as the sun reaches its zenith during noon, the shaded ground area underneath the array is minimum, and thus, the R_A^* is higher owing to a higher ground-reflected irradiance and resulting higher I_{rear}. When compared with Fig. 6(a) and (c), it is not surprising that the increase in R_A leads to an increase in the R_A^* as well; however, for a farm configuration, we find that R_A^* is always less than R_A throughout the day. This is a defining property of solar farms: closely packed, the periodic rows of module arrays in a farm result in a *lower* R_A^* compared with a solitary/standalone array ($p/h \to \infty$). Since the R_A^* and I_{front} variables do not depend on the technology (HIT or tandem), we obtain identical traces for the same albedo conditions [cf., Fig. 6(b) and (e)]. However, of course, the corresponding G values depend on the technology.

At low albedo [$R_A = 0.1$, Fig. 6(a)], the 3/4T-tandem cell demonstrates a higher G (~20% near midday) during each day compared with the 2T-tandem cell. This is because, for this albedo condition, the 2T-tandem's efficiency was drastically lowered, owing to a smaller bottom-cell current resulting from low rear irradiance. By increasing albedo to $R_A = 0.3$ [see Fig. 6(b)], 2T-tandem's efficiency improves, but 3/4T tandem still outperforms (7% gain in YY) because of its lower sensitivity to the variation in R_A^*. Fig. 6(c) shows a high-albedo scenario when $R_A = 0.8$. Under such conditions, the 2T-tandem's performance will be again affected due to the current being limited by the perovskite top cell. As a result, in such albedo conditions, 4T will outperform 2T tandem by a larger margin (∼8% gain in YY). Similar patterns can also be observed in regions experiencing predominantly diffuse sunlight: for example, Fig. S3 in the supplementary material shows the R_A^* characteristics for different R_A in Nanjing, China. A higher diffuse fraction in Nanjing results in enhanced rear irradiance and R_A^*, leading to improved yield potentials when the albedo is low. These improvements are evident from the reduced gains of 3/4T compared with 2T at locations primarily exposed to diffuse light, as depicted in Fig. 4(a). However, despite these enhancements, the discrepancy persists between R_A^* required to achieve current matching between subcells (as determined by cell design) and actual R_A^* obtained in the farm, as shown in Fig. S3.

The figures on the bottom row of Fig. 6 illustrate the relative cell efficiencies of 3/4T tandem to its constituent HIT cell. At each albedo condition, we see that the 3/4T tandem is the preferred option for maximizing energy yield at this location. Especially, if the albedo is low ($R_A = 0.1$), the 3/4T tandem should be preferred to realize high gains in yield. Given the 2T tandem's significantly diminished performance at such low albedo conditions, HIT will be the preferred second option for deployment. These conclusions are generally valid for all the locations in the world unless the bifacial 2T-tandem cells were explicitly designed for different albedo conditions.

It is important to note that the smooth R_A^* traces, as depicted in Fig. 6, reflect our assumption of a prototypical clear-sky day. This "time-averaged" approach determines the yearly energy yield accurately [38], although the daily energy yield, based on actual, time-dependent direct/diffuse irradiance profile, would yield much noisier R_A^* traces due to cloud movements.

To summarize, it was demonstrated that R_A^* directly determines the current-matching condition in bifacial 2T-tandem cells and its relative benefits. In a solar farm, R_A^* is time varying. Consequently, variations in R_A^* result in time-varying mismatches in subcell currents throughout the day for 2T-tandem cells. Such cell-level current mismatch and associated loss present additional challenges in terms of reliability for bifacial 2T-tandem cells. However, by electrically decoupling the subcells in 3/4T architecture, the impact of time-varying effective albedo is avoided.

V. Summary and Conclusion

In this study, we have presented the global performance potential of solar farms utilizing bifacial 3/4T PVK-Si tandem solar cells for the first time. The potential yields and advantages of 3/4T-tandem cells were calculated for three values of $R_A : 0.1$, 0.3, and 0.8. The maps can be used in two ways: one can use the natural R_A for a particular location and interpolate from the maps provided to see the expected benefits. Alternatively, one can determine the benefits of engineering surface R_A to specific values.

In computing the results presented above, several assumptions have been made. First, we disregarded any parasitic optical losses, treating the 3T and 4T configurations as electrically equivalent. Additionally, losses related to cell-to-module upscaling were neglected. To reduce the computational burden, location-specific irradiance spectrum, and surface-specific spectral albedo have been neglected. In practice, time-varying spectral irradiance and R_A will lead to additional variability in yield potential. Indeed, this additional variability would make the gain potential of 2T tandem at suboptimal albedo even more uncertain. Therefore, the location-specific spectral issues must be considered in future works on this topic, especially for considering the 2T-tandem performance for specific farm locations.

The key takeaways of this paper are as follows.

1) Time-varying effective albedo (R_A^*) determines the ultimate efficiency of bifacial tandems. The current-matching requirement in bifacial 2T architecture entails that its performance benefits will largely depend on R_A^* when deployed. Consequently, for bifacial 2T-tandem cells to yield substantial benefits over single-junction cells, it becomes necessary to purposefully design them for different albedo conditions, making them adaptable across various geographical regions. As shown above, to maximize the benefits of a 2T-tandem cell designed for $R_A^* = 0.18$, R_A of HSAT solar farm would have to exceed 0.18.
2) Time-varying R_A^* in a solar farm and its disparity with R_A^* required for current matching entails continuous current mismatch in 2T-tandem cells when deployed. This constant mismatch in current introduces additional heating, which further presents reliability challenges for bifacial 2T-tandem cells.
3) Bifacial 3/4T configurations do not suffer from the current-matching issue under variable R_A^* conditions. As such, the 3/4T configuration allows for greater flexibility in terms of subcell design and deployment in various albedo conditions.
4) 3/4T tandem is predicted to outperform the 2T tandem and its constituent HIT cell, respectively, by an average of 5% and 23% gains when $R_A = 0.3$. Although there may be some reduction in these gains due to additional losses in practical applications, they still represent a significant improvement over single-junction silicon cell. This substantial increase in performance has the potential to compensate for the higher manufacturing costs associated with 3/4T-tandem technology in high-performance and high-value markets. However, a comprehensive technoeconomic study will be required to fully assess the economic feasibility of implementing 3/4T-tandem cells. This analysis, which takes into account realistic solar farm conditions, serves as the initial step toward conducting such a study.
5) The performance of standalone bifacial systems does not provide the accurate indication of the relative advantage of bifacial PV technology when it is deployed in a solar farm due to the significant shading effect between rows. To demonstrate the performance benefits specific to the technology when deployed in a solar farm, it is necessary to compare different technologies, such as the PVK-HIT tandem and the constituent HIT, under identical configurations, whether it be tracking or fixed tilt. This study

follows this approach to assess the technology-related performance benefits.

Given the substantial energy yield potential of bifacial 3/4T-tandem technologies, our results should inspire and guide further research related to issues related to scaling the 3/4T tandem to system scale to finally realize the promise of next-generation high-efficiency PV systems.

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